- **Deficiency:** Deficiency of protein causes protein energy malnutrition which covers a wide spectrum of clinical stages ranging from the severe forms like kwashiorkor and marasmus to the milder forms like growth retardation. Protein energy malnutrition is due to "food gap" between the intake and requirement. The average energy deficit in Indian children is 300kcal/ day. Deficiency of protein is discussed in detail in the section protein energy malnutrition



▲ Butter



▲ Ghee

5.2.3 Lipids/Fats

- **Definition:** Lipids are organic substances soluble in fat solvents such as alcohol, ether, and chloroform but not in water. The term includes fatty acids, soaps, neutral fats, phospholipids, steroids and waxes. Oils found in seeds, butter from milk, and lard from meat, are examples of fats found in foods.

Table 1 Classification of Lipids

Simple Lipids	Compound Lipids	Derived Lipids
Fats and Oils	Phospholipids	Hydrolytic substances of simple and compound lipids
Waxes	Glycolipids	
	Lipoprotein	

Source: Swaminathan.M 2012, Foods and Nutrition, Bangalore Printing and Publishing Company, Bangalore.

- **Classification:** Lipids are classified as follows
- **Functions:** The functions of lipids are to
  - Provide energy reserve and supply 9 kcal/gram.
  - Serve as a vehicle for the absorption of fat-soluble vitamins A, D, E, and K
  - Supply essential fatty acids necessary for growth and function
  - Provide energy source so that proteins are spared for tissue growth and repair
  - Gives satiety
  - Act as insulators against heat and cold.
- **Food Sources:** Visible fat sources are Butter, Ghee and Oil. Invisible fat sources are Cereals, Pulses, Oil seeds, Milk and Egg.

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